
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field, obtained by differentiating a trained decoder and validated against analytic
6 truth in that regime, finds mean-curvature magnitude correlating negatively with
7 cross-survey neighbour agreement at every latent dimension, and that local den-
8 sity removes most of it at $d = 20$, half at $d = 25$ and little of the already small
9 $d = 32$ value. A second-order expansion of a linear map restricted to the manifold
10 explains why: a readout sees the second fundamental form contracted with its nor-
11 mal component, $\langle w_N, \Pi \rangle$, and, at second order, an alignment’s error splits into a
12 connection mismatch and the relative second fundamental form projected into the
13 target normal space; mean-curvature magnitude sees neither at large codimension.
14 On a unit-normalized embedding that Hessian splits into a sphere term, propor-
15 tional to the readout’s own value, and a shape term. On a synthetic surface the
16 sphere term has partial correlation -0.74 to -0.80 with local R^2 across samplings
17 that swing the $\|H\|$ partial from -0.29 to $+0.26$; on 86,471 labelled galaxies it
18 is negative on every label, the mismatch between the label’s own Hessian and the
19 shape term is negative wherever the label has local structure, and the shape term
20 alone takes the label-dependent sign that mismatch predicts. Mean curvature of a
21 single representation is therefore not a viable explanation for weak Platonic conver-
22 gence; the obstruction is relative and, for the fitted global linear alignment, already
23 present at first order.

24 1 Introduction

25 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
26 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
27 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
28 a family of different vision models share only 0.3 to 2.3 percent of their nearest neighbours across
29 the two surveys. We ask where on the manifold that agreement lives: if two models disagree most
30 where the embedding surface bends sharply, local mean curvature should associate negatively with
31 local cross-survey agreement. We test this claim using a pointwise mean-curvature estimation from
32 a trained autoencoder on the embedding space.

33 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
34 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
35 test of the curvature–alignment prediction under two probes, which finds no effect free of density;

(iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map responds to and shows mean-curvature magnitude is blind to it; and (iv) known-surface and real-data tests on the readout side, where the expansion’s terms behave as predicted.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. Embeddings are unit-normalized (data on S^{767}). Persistent homology detects no one-dimensional cycles ($\beta_1 = 0$ under bootstrap bands [Fasy et al., 2014]); the auto-encoder is a computational chart, and curvature is evaluated only where J has full rank.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector is $H = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape. To account for this, the map differentiated is $F/\|F\|$ (after which $H_{\text{rad}} = -d$ to 10^{-14}) and $\|H_{\text{tan}}\|$ carries the shape. Decoder curvature has served as a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is an instrument, validated against analytic fixtures.

Validation. Scored against closed-form curvature on rank (Spearman), direction (cosine) and magnitude ratio. At $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H_{\text{tan}}\|$ (equivalently $\|H\|$, the radial part being fixed), direction cosine 0.999 and magnitude ratio 1.00 for the full vector, and rank 0.75 under noise at 25% of patch radius.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local, correlated with $\|H\|$ over all 10,000 points at $d \in \{20, 25, 32\}$. Centered kernel alignment [Kornblith et al., 2019] is the second probe.

Result. The raw association $\rho(\|H\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-sensitive [Radovanović et al., 2010]: density correlates with agreement at -0.21 at every d and with $\|H\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 : 78%, 49% and 8% of the association removed. Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; replicating across three decoder seeds; the CKA tertile gap within strata agrees at $d = 20$ and 25 , -0.05 to -0.08 against a null band of ± 0.017). Replacing $\|H\|$ by its sphere-tangential part $\|H_{\text{tan}}\|$ (the field here is the unprojected decoder’s, radial part not fixed; the two rank at $\rho = 0.89$ – 0.96) collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$,

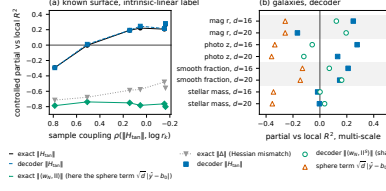


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample’s density–curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder’s $\|H_{\text{tan}}\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

83 SO

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

84 To second order, a linear function is as non-linear on M as M bends in that function’s normal direc-
 85 tion [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the
 86 rest, $\Pi = \Pi^S - g \hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$ (2): a linear readout of a unit-normalized
 87 embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion
 88 to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a
 89 patch $u \sim N(0, s^2 I_d)$ in normal coordinates (the $\frac{1}{2}$ below uses Gaussian fourth moments), with $\bar{c}$
 90 the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

91 For two embeddings F, G of the same objects in a common chart u and a linear map A with
 92 $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G -$
 93 $A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear align-
 94 ment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter
 95 when the induced connections agree at x_0 (in particular under a local intrinsic isometry); the extrinsic
 96 term’s normal part is $\Pi_G - P_N^G A \Pi_F$.

97 So the curvature a linear readout pays for is $\langle w_N, \Pi \rangle$, and an alignment’s second-order error has
 98 an extrinsic part whose normal component is $\Pi_G - P_N^G A \Pi_F$; $\|H\|$ bounds only a trace and need
 99 not correlate with either at codimension 750; the term carries s^4 , so patch radius, hence density,
 100 multiplies it; and when the label has intrinsic curvature of its own, Δ carries the residual and the
 101 curvature piece can take either sign.

102 5 The probe-facing curvature, on a known surface and on galaxies

103 The readout side can be tested directly: w is known and Π comes from the decoder. Statistic: five-
 104 fold out-of-fold ridge probe [Alain and Bengio, 2016] at $\alpha = 100$ from embedding to label; local
 105 R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with
 106 Freedman–Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label
 107 variance and count and, as a stronger control, log radius at $k \in \{16, 64, 256, 1024, 2048\}$. Per anchor
 108 we compute $\|H_{\text{tan}}\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by
 109 Eq. (2) into its shape and sphere terms.

110 **Known surface.** The validation surface is resampled with weight $\|H_{\text{tan}}\|^\gamma$, so only where the points
 111 sit changes (Figure 1a). The $\|H_{\text{tan}}\|$ partial follows the sampling from -0.29 at coupling $+0.80$
 112 through zero to a plateau near $+0.22$, and the decoder reproduces it. The probe-facing Hessian
 113 $\|\langle w_N, \Pi \rangle\|$ has partial correlation -0.74 to -0.80 at every sampling. On this surface it is the sphere
 114 term of Eq. (2): rank 0.996 – 1.000 with $\sqrt{d}|\hat{y} - b_0|$, the in-sphere part 20 to 40 times smaller. The
 115 fixture thus confirms the expansion’s sphere term: an intrinsically linear label loses local R^2 where
 116 the readout’s own value is large, whatever the sampling. With a nonlinear label the sphere term is
 117 indistinguishable from zero and $\|\Delta\|$ carries the partial (-0.2 to -0.6), as Eq. (3) predicts.

118 **Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing
 119 Hessian (rank 0.96 – 0.99 with the full norm). To read it through Eq. (3) we estimate $\text{Hess}_M y$ per

label	$\ H_{\text{tan}}\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours, where, at the anchor, the Christoffel symbols vanish to the needed order and the quadratic coefficient is the covariant Hessian. Three regularities (Table 1). The sphere term is negative on every label and both d (-0.13 to -0.36), as on the fixture. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$); local R^2 is higher where the manifold bends in the readout’s direction the way the label bends. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift), consistent with Eq. (3): it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$, so raising it can shrink or enlarge the mismatch depending on its alignment with the label Hessian; empirically its sign follows the cross term’s where that is nonzero and the remaining square where it is null. $\|H_{\text{tan}}\|$ changes sign between $d = 16$ and 20 for magnitude: it sits on the curve of Figure 1a (coupling to radius -0.60). The same regression applied to the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median metric cosine with the decoder’s 0.55 – 0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace, run on five encoders at the same 512 anchors ($d = 16$, $k = 2,048$: ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L), gives density-controlled associations with the magnitude probe’s local R^2 of -0.24 , 0.00 , $+0.27$, $+0.23$ and -0.19 . A trace statistic has no encoder-stable sign, as Eq. (1) predicts; a cross-encoder test should carry the terms of Eq. (3).

6 Discussion

A decoder yields a pointwise second fundamental form at production scale whose readout-facing component agrees with an independent estimator and with the data. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the sphere term and the Hessian mismatch carry the predicted negative sign, and the shape term is label-signed because it enters through its alignment with the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. MKNN and CKA are invariant to rigid motions of either embedding, so if weak convergence is attributed to the failure of a linear alignment map, one embedding’s mean curvature is not the relevant second-order quantity; after tangent alignment the relevant normal term is $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot at $d = 20$ (2,048 anchors) supplies both and finds, for the fitted global ridge alignment, substantial mismatch already at first order: it carries HSC tangent frames into the Legacy embedding with 44% residual, and the second-order term shows no density-free association yet ($+0.04$, n.s.; alternative estimators -0.24 to $+0.13$). In sum, mean curvature of a single representation does not explain weak Platonic convergence; the obstruction is relative and, for the fitted linear alignment, already first order; testing it means aligning tangent frames first and then the relative second fundamental form, for which the instrument is in hand.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10 – 0.17); a same-size fit in the quadratic-chart estimator’s frame failed a split-half gate at cosine 0.20 . The sphere term is not separated from ridge shrinkage of extreme predictions. Permutation p -values assume anchor exchangeability under overlapping patches.

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